

$$C = \frac{Q}{V}$$

$$V_{BA} = V_B - V_A$$

$$V = \frac{Q}{C}$$

A) Serie

$$C_{eq} = \frac{Q}{V_A - V_B}$$

$$C_1 = \frac{Q}{V_A - V_M}$$

$$C_2 = \frac{Q}{V_M - V_N}$$

$$C_3 = \frac{Q}{V_N - V_B}$$

$$\begin{aligned} V_A - V_M &= \frac{Q}{C_1} \\ V_M - V_N &= \frac{Q}{C_2} \\ V_N - V_B &= \frac{Q}{C_3} \end{aligned}$$

$$\begin{aligned} V_A - V_B &= \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} \\ V_A - V_B &= Q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right) \end{aligned}$$

$$\frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}} = \frac{Q}{V_A - V_B} \Rightarrow$$

$$C_{eq} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots}$$

B)

$$C_1 = \frac{Q_1}{V_A - V_B}$$

$$C_2 = \frac{Q_2}{V_A - V_B}$$

$$C_3 = \frac{Q_3}{V_A - V_B}$$

$$\begin{cases} Q_1 = C_1 (V_A - V_B) \\ Q_2 = C_2 (V_A - V_B) \\ Q_3 = C_3 (V_A - V_B) \end{cases} \Rightarrow$$

$$Q = Q_1 + Q_2 + Q_3$$

$$C_e = \frac{Q}{V_A - V_B}$$

$$\Rightarrow Q = Q_1 + Q_2 + Q_3$$

$$Q = C_1 (V_A - V_B) + C_2 (V_A - V_B) + C_3 (V_A - V_B)$$

$$Q = (V_A - V_B) (C_1 + C_2 + C_3)$$

$$C_1 + C_2 + C_3 = \frac{Q}{V_A - V_B} \Rightarrow$$

$$C_{eq} = C_1 + C_2 + C_3$$

$$C = \frac{Q}{V}$$

$$W_{el} = \begin{cases} \frac{1}{2} Q V_{AB} \\ \frac{C \cdot V_{AB}^2}{2} \\ \frac{Q^2}{2C} \end{cases}$$

$\rightarrow$ dem.